

MD. REJAUL KARIM

Senior Software Engineer — Native Android | Kotlin · Jetpack Compose · Hardware Systems · Flutter

Dhaka, Bangladesh · +880 1724821030 · rejaul.karim.pro@gmail.com · LinkedIn · Portfolio · GitHub

PROFESSIONAL SUMMARY

Senior Software Engineer with 7+ years delivering production-grade Native Android systems across ePOS/KDS commercial hardware platforms, EdTech at scale (100K+ MAU), and FinTech on-device ML pipelines. Core expertise spans low-level hardware IPC via AIDL, ESC/POS thermal rendering on Sunmi and Rockchip commercial hardware, enterprise .aar SDK modularization, and Jetpack Compose architecture. Measurable delivery record: crash-free sessions lifted from 91% to 98.87%, 25% KYC onboarding conversion uplift, 19% booking conversion on a 360K+ user platform. Combines systems-level hardware depth with rigorous Clean Architecture discipline to ship reliable, maintainable software at sustained velocity.

TECHNICAL SKILLS

Languages: Kotlin, Java, Dart

Native Android: Jetpack Compose, Material 3, Coroutines, Flow, StateFlow, SharedFlow, ViewModel, Navigation Component, WorkManager, Lifecycle, Paging 3, CameraX, ViewBinding

Architecture & DI: MVVM, Clean Architecture, Repository Pattern, SOLID, Modularization, Hilt

Hardware & Embedded POS: ESC/POS Protocol, Sunmi SDK (NT311, D3 Pro, Cloud/Label Printers), Rockchip ctd-rk3288, AIDL (Inter-Process Communication), Bluetooth / USB / LAN / Wi-Fi Thermal Printing, Real-time Order Routing, Offline-first Sync

SDK & Library Engineering: .aar Modular SDK Architecture, Gradle Product Flavors, Multi-app SDK Distribution

Flutter / Cross-Platform: Flutter, Dart, Riverpod, BLoC, Cubit, Freezed, Dio, Hive

Networking & APIs: Retrofit, OkHttp, Ktor, REST, WebSocket, Interceptors, Token Refresh, Pagination

Persistence: Room, SQLite, ObjectBox, DataStore, SharedPreferences, Hive

AI / ML: ML Kit (Face Detection, OCR, NID Verification), Prompt Engineering, Cursor, Claude Code, GitHub Copilot

CI/CD & Release: GitHub Actions, Firebase App Distribution, Gradle (KTS), Google Play Console, ProGuard / R8

Analytics & Monitoring: Firebase Crashlytics, Firebase Analytics, Mixpanel, WebEngage, Sentry, LeakCanary

PROFESSIONAL EXPERIENCE

Senior Software Engineer (App Development)

VALT | Sylhet, Bangladesh

Feb 2026 – Present

- Led the XML-to-Jetpack-Compose migration of the KDS application UI layer, introducing a component-based architecture with dynamic screen scaling and runtime theme switching — reducing UI-related maintenance overhead and enabling faster iteration across different commercial display hardware.
- Built a shared thermal printer .aar library used across the ePOS and KDS apps, handling ESC/POS command generation and communication over Bluetooth, USB, and LAN/Wi-Fi — standardizing printer integration and removing duplicated driver code across the project.
- Implemented a background print service using AIDL to manage print jobs off the main thread, resolving a crash and state corruption issue that occurred when multiple orders were submitted simultaneously on Sunmi NT311 and D3 Pro hardware.
- Built the 80mm receipt rendering pipeline for Rockchip ctd-rk3288 hardware, handling ESC/POS byte formatting and print buffer management to produce consistent, low-latency output under high order volume.
- Refactored legacy Java API calls scattered across the ePOS and KDS modules into a unified Kotlin-based networking layer, standardizing error handling and removing redundant call sites throughout the codebase.
- Built the order routing and live ticket display logic for the KDS app using Kotlin Coroutines and Flow, ensuring kitchen tickets update promptly and accurately across multiple display stations during peak service.

Senior Mobile Application Developer

Repocket | Singapore (Remote)

Nov 2025 – Jan 2026

- Engineered the Referral and Rewards feature end-to-end in Flutter / Dart — validated by a 15% user engagement uplift within the first month of launch — by designing a deep-link-aware referral graph with server-synchronized reward state and idempotent claim logic that eliminated reward duplication under network-retry conditions.
- Reduced funnel abandonment across referral steps by instrumenting targeted analytics events to surface per-step conversion drop-offs, producing behavioral data that directly informed two post-launch product iterations.
- Delivered consistent cross-platform reliability across Android and iOS through widget-level testing, Dart DevTools profiling, and code review within a distributed international engineering team.

Specialist, Engineering — Android

10 Minute School | Dhaka, Bangladesh

Aug 2021 – Jun 2025

Promoted from Associate Software Engineer · Aug 2021 – May 2023

- Lifted crash-free session rate from 91% to 98.87% — recovering 7.87 percentage points of session reliability for a 100K+ MAU platform — through systematic root-cause triage via Firebase Crashlytics, exception-handling hardening at identified failure boundaries, and proactive pre-release regression monitoring.
- Drove full XML-to-Jetpack-Compose migration of the core EdTech application UI layer, accelerating new feature throughput by 20%, by eliminating imperative view-binding overhead, phasing out deprecated transition APIs (including overridePendingTransition), and establishing a lifecycle-correct, recomposition-aware component model.

- Architected a Compose-based design system comprising 30+ reusable, accessibility-compliant UI components, cutting design-to-dev handoff cycles and reducing accumulated UI design debt by 60% across all active product surfaces.
- Delivered end-to-end ownership of the TARA adaptive learning app (Monash University research collaboration) — including a spaced-repetition content delivery engine, adaptive assessment layer, and early childhood literacy/numeracy interaction model — from architecture through Play Store release.
- Led Java-to-Kotlin migration of the monolithic Android codebase, improving null-safety guarantees, reducing incremental build times, and establishing idiomatic Kotlin patterns adopted as the engineering standard across all subsequent feature development.
- Engineered the booking system — achieving 19% conversion on a 360K+ registered user base and lifting class attendance from 16% to 35% — by architecting optimistic local state, WebSocket-driven real-time availability updates, and transactional rollback logic on reservation failure.
- Built core EdTech modules from scratch — live video classrooms, real-time chat, personalized learning flows, and authentication — serving 100K+ active users with WebSocket-backed communication and resilient session state management.
- Engineered an interactive exam module that improved student retention by 23% and gamified learning flows that increased engagement by 12%, through real-time scoring feedback, adaptive question sequencing, and streak-aware mechanics in Jetpack Compose.

Assistant Software Engineer — Android

Millennium Information Solution Ltd. | Dhaka, Bangladesh

Oct 2020 – Jul 2021

- Engineered a reusable .aar SDK integrating ML Kit on-device facial recognition, document OCR, and NID field verification — distributed across three FinTech banking apps (Social Islami Bank, Union Bank, Al-Arafah Islami Bank) via Gradle product flavors — reducing per-app KYC integration overhead from weeks to hours.
- Automated the full KYC onboarding pipeline using on-device ML inference (face liveness detection, document OCR, NID extraction), increasing average account creation rates by 25% across all three banking applications.
- Implemented self-registration flows delivering 50+ verified weekly activations per app, by engineering multi-step form logic with inline validation, session-persistent state, and CameraX-backed document capture.
- Sustained high-compliance, zero-incident monthly feature delivery across three concurrent Android banking applications within Agile/Scrum cycles.

Junior Software Engineer — Android

CodeNext Ltd. | Dhaka, Bangladesh

Apr 2019 – Oct 2020

- Architected and shipped three B2B2C Android applications — Ichiiba wholesale marketplace, Salesman mobile app, and a tablet-optimized POS system — engineering a centralized API integration layer via Retrofit and OkHttp with request/response interceptors, transparent token refresh, and structured error-handling contracts.
- Designed the tablet-optimized POS application with offline transaction queuing and multi-SKU barcode scan workflows, establishing early POS architecture expertise foundational to subsequent ePOS system engineering.
- Published Ichiiba wholesale marketplace to Google Play Store, integrating Firebase Cloud Messaging for in-app re-engagement across a B2B2C multi-stakeholder user base.

Android Developer

Digital Internet Marketing | Dhaka, Bangladesh

Apr 2018 – Oct 2018

- Delivered a production Android car rental application as first-career production software, engineering real-time booking state synchronization via Firebase Realtime Database and FCM-driven push notifications for low-latency booking management.

SELECTED PROJECTS

- **10 Minute School** — Native Android EdTech platform, 100K+ MAU. Live classes, real-time chat, gamified learning. Kotlin, Compose, MVVM, Room, Retrofit. [\[Play Store\]](#)
- **TARA** — Adaptive learning app for early childhood literacy and numeracy, Monash University collaboration. Full Android ownership. [\[Play Store\]](#)
- **Repocket App** — Cross-platform Flutter app (Android + iOS) for passive income. Referral and Rewards feature, +15% user engagement. [\[Play Store\]](#)
- **aib i-Banking** — Secure FinTech banking app with ML Kit on-device KYC pipeline. +25% account creation across three banking apps. [\[Play Store\]](#)
- **SIBL Now** — Internet banking Android app for Social Islami Bank with secure authentication and fund transfer. [\[Play Store\]](#)
- **Ichiiba Wholesale Marketplace** — B2B2C Android marketplace platform. Java/Kotlin, Retrofit, FCM, Room. [\[Play Store\]](#)

EDUCATION

Bachelor of Science, Computer Science & Software Engineering

2013 – 2018

American International University Bangladesh (AIUB) | Dhaka, Bangladesh | CGPA: 3.36 / 4.0

Relevant Coursework: Data Structures & Algorithms, OOP, Software Architecture, Database Systems, Discrete Mathematics

CERTIFICATIONS

- Introduction to Prompt Engineering — Simplilearn (2025) [\[View\]](#)
- Advanced Android Application Development (Score: 82%) — Bangladesh Digital Skills / BDSkills (2024) [\[View\]](#)
- Top-up IT Training on Android Development — Bangladesh Computer Council (BCC) / ICT Division, validated by George Washington University (2018) [\[View\]](#)

LANGUAGES

Bengali (Native) · English (Professional Working — C1 Listening/Writing, B2 Speaking) · Hindi/Urdu (Limited Working)